Malik Weren

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Education

ETH Zurich Sept 2025 – Aug 2027
M.Sc. Electrical Engineering and Information Technology *Zurich, Switzerland*
– Specializing in **electronics and machine learning**; coursework in computer architecture, digital circuit design, deep learning, bayesian learning, RL, SoC for ML, HW/SW codesign, Neuromorphic

Eindhoven University of Technology (TU/e) Aug 2022 – July 2025
B.Sc. Electrical Engineering + Entrepreneurship Program *Eindhoven, Netherlands*
– Graduated **cum laude**, GPA 8.2/10 (Top 10%)

Technical Skills

Hardware & Physical Design: Computer architecture, logic design, arithmetic units, synthesis, place & route

Digital Design & FPGA: SystemVerilog, RTL design, system design, transistor level circuits

EDA / ASIC Tools: Yosys, OpenROAD, Cadence Virtuoso, Cadence Innovus, Synopsys, Vivado, Ngspice

Programming & ML: Python, PyTorch, TensorFlow, C, MATLAB, scikit-learn

ML for Hardware: Hardware Design-space exploration, PPA optimization, TinyML, model compression & quantization

Experience & Research

Semester Thesis – Efficient arithmetic cores for AI Hardware Accelerators Feb 2026 – June 2026
Huawei Zurich Research Center & ETH Zurich *Zurich, Switzerland*
– Design-space exploration of various variants of matrix multiplication cube GEMM architectures for transformer workloads
– Implementing a pipeline for high-effort synthesis via **Yosys + OpenROAD** and commercial tools to improve PPA
– Evaluating PPA trade-offs; preparing top candidates with retiming for **test-chip tapeout** to validate silicon improvements
– Building automated flows bridging design generators, open source chip tools, and physical commercial tool flow

Bachelor's Thesis – ML-Enhanced SAR ADC Post-Correction Feb 2025 – July 2025
Integrated Circuits Group, Eindhoven University of Technology *Eindhoven, Netherlands*
– Developed a novel **neural-network-based post-correction architecture** for SAR ADC, combining ML with analog IC circuit
– Targeting publication at **IEEE AI-circuits conference**, but put on pause due to masters obligations

AI Tech Consultant (National Honors Program) Mar 2025 – Aug 2025
Royal De Heus *Eindhoven, Netherlands*
– Built ML models and an AI-powered forecasting dashboard for data-driven supply chain decisions

Electrical Engineer June 2024 – Dec 2024
VOID Rocketry Team *Eindhoven, Netherlands*
– Co-developed power electronics PCB for rocket avionics; implemented embedded C/C++ for sensor interfaces

Teaching Assistant Nov 2023 – Present
TU/e; ETH Zurich
– Led tutorials and mentoring for 100+ students across 10+ EE/CS courses

Selected Projects

MX-Format KV Cache Compression & GEMM Acceleration for LLM Inference – *ETH Zurich, IIS (Prof. Benini)*
– Analyzed KV cache compression strategies (**MX quantization**, Huffman coding) for MobileLLM inference; benchmarked MX-format workloads on the **MXCore GEMM accelerator** integrated with the RISC-V Snitch cluster

TinyML – Machine Learning on Microcontrollers
– Optimized a ResNet model for MCU via quantization, pruning, and architecture-aware compression; achieved 100× size reduction with 1–2% accuracy drop and fast inference through hardware acceleration insights

Probabilistic AI & Decision-Making Under Uncertainty | *ETH Zurich*
– Implemented GP regression with uncertainty quantification, calibrated Bayesian neural networks (SWAG), risk-aware Bayesian optimization with safety constraints, and off-policy reinforcement learning (TD3) for control

Neural State Machines on Neuromorphic Hardware (DYNAP-SE) | *INI508, UZH/ETH*
– Deployed attractor-based spiking neural state machines on mixed-signal DYNAP-SE chip for sequential memory replay and rhythm encoding